Product Analytics Dashboard Documentation

Project Overview

The Product Analytics Dashboard is an end-to-end Business Intelligence project developed to analyze product performance, sales trends, pricing behavior, category contribution, and product activity using PostgreSQL and Tableau.

This project simulates a real-world Product Analytics workflow where raw transactional product data is transformed into meaningful business insights through SQL analytics and interactive dashboards.

The dashboard helps business stakeholders understand:

- Which products generate the highest revenue
- Which categories dominate sales
- Product pricing performance
- Product demand behavior
- Product segment profitability
- Product sales trends over time

Project Objectives

The main objectives of this project were:

- Analyze product sales performance
- Identify top-performing products
- Understand category-level revenue contribution
- Measure product pricing trends
- Track product sales growth over time
- Analyze product activity using recency analysis
- Build an executive-level Product Analytics Dashboard

Technology Stack

Component	Technology
Database	PostgreSQL
Query Language	SQL
Visualization	Tableau
Analytics	Product Analytics
Data Modeling	Star Schema Concepts

Component	Technology
Data Warehousing	Bronze, Silver, Gold Layers

Data Warehousing Concepts Applied

This project follows modern layered data warehouse architecture.

Bronze Layer

Purpose

Store raw source data without transformations.

Implemented Concepts

- Raw product transaction ingestion
- Source-level data storage
- Full data loading
- Data traceability

Silver Layer

Purpose

Data cleaning and transformation layer.

Implemented Concepts

- Data standardization
- Missing value handling
- Derived metrics generation
- Product enrichment

Gold Layer

Purpose

Business-ready analytical reporting layer.

Implemented Concepts

- KPI-ready datasets
- Product reporting tables
- Aggregated analytical measures
- Dashboard-ready views

ETL & Data Processing Workflow

The project simulates real-world ETL operations using SQL workflows.

Implemented operations include:

- Bulk data loading
- Data transformation
- Data cleansing
- Aggregation logic
- Analytical dataset creation
- Reporting layer preparation

Dashboard Overview

The Product Analytics Dashboard contains:

1. KPI Cards
2. Product Revenue Trend
3. Revenue by Product Segment
4. Sales by Category
5. Top Products by Revenue
6. Product Recency Analysis
7. Average Selling Price by Category

KPI Section

KPI 1 — Total Sales

What It Measures

Total revenue generated from all product sales.

Business Question Answered

“How much revenue was generated from products?”

Value Observed

29.35M

Business Insight

The company generated approximately $29 million in product revenue, indicating strong product demand and healthy business performance.

KPI 2 — Total Products

What It Measures

Total number of unique products available in the business.

Business Question Answered

“How many products does the business manage?”

Value Observed

130

Business Insight

The business maintains a moderately sized product portfolio, enabling focused product performance analysis and category optimization.

KPI 3 — Total Quantity

What It Measures

Total quantity of products sold.

Business Question Answered

“How many product units were sold overall?”

Value Observed

60,404

Business Insight

High sales volume indicates strong market demand and active customer purchasing behavior.

KPI 4 — Average Selling Price

What It Measures

Average selling price across products.

Business Question Answered

“What is the average selling price of products?”

Value Observed

142,331

Business Insight

The business operates with relatively high-value products, indicating premium pricing and strong revenue generation per product.

Visualization Analysis

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1. Product Revenue Trend

Chart Type

Line Chart

Purpose

Analyze product revenue growth over time.

Business Questions Answered

- Is product revenue increasing over time?
- Which years generated the highest revenue?
- Are there major revenue fluctuations?

Key Insights

- Revenue declined slightly during early years
- Strong growth observed during 2013
- Revenue decreased again afterward
- Indicates periods of strong product demand followed by decline

Business Interpretation

The business experienced significant product growth during 2013, possibly due to:

- Increased market demand
- Product expansion
- Successful sales campaigns
- High-performing product launches

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2. Revenue by Product Segment

Chart Type

Horizontal Bar Chart

Product Segments

- High-Performer
- Mid-Performer
- Low-Performer

Purpose

Compare revenue contribution across product segments.

Business Questions Answered

- Which product segment generates the highest revenue?
- Which segment drives business profitability?

Key Insights

- High-Performer products dominate revenue contribution
- Mid-Performer products contribute moderate revenue

- Low-Performer products contribute minimal revenue

Business Interpretation

A small group of high-performing products contributes the majority of business revenue.

Possible business actions:

- Focus marketing on high-performing products
- Optimize inventory for top products
- Re-evaluate low-performing products

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3. Sales by Category

Chart Type

Horizontal Bar Chart

Categories

- Bikes
- Accessories
- Clothing

Purpose

Analyze revenue contribution by product category.

Business Questions Answered

- Which category generates the most revenue?
- Which product category drives business growth?

Key Insights

- Bikes overwhelmingly dominate sales revenue
- Accessories contribute significantly less
- Clothing contributes the lowest revenue

Business Interpretation

The business is highly dependent on bike sales.

Possible business implications:

- Bikes are the primary revenue driver
- Accessory and clothing categories may need optimization

- Opportunity exists for cross-selling accessories

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4. Top Products by Revenue

Chart Type

Horizontal Bar Chart

Purpose

Identify highest revenue-generating products.

Business Questions Answered

- Which products generate maximum revenue?
- Which products should receive business focus?

Key Insights

- Mountain-series bikes dominate top revenue positions
- Road-series bikes also contribute strongly
- Revenue concentration exists among a small number of products

Business Interpretation

Premium bike products are the strongest business performers.

Possible business actions:

- Prioritize inventory for top products
- Improve marketing around high-performing models
- Develop premium product strategies

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5. Product Recency Analysis

Chart Type

Histogram

Purpose

Analyze product activity using recency behavior.

What Recency Means

Recency measures:

“How many days ago the product was last sold.”

Lower recency:

- Recently active products

Higher recency:

- Less active or slow-moving products

Business Questions Answered

- Are products actively selling?
- Which products are becoming inactive?
- How healthy is product movement?

Key Insights

- Majority of products fall within lower recency ranges
- Most products remain recently active
- Only a small number of products show higher inactivity

Business Interpretation

The business maintains strong product movement and healthy sales activity across most products.

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6. Average Selling Price by Category

Chart Type

Horizontal Bar Chart

Purpose

Compare pricing behavior across product categories.

Business Questions Answered

- Which category has the highest average selling price?
- Which products are premium-priced?

Key Insights

- Bikes have significantly higher average selling price
- Clothing and accessories remain lower-priced
- Pricing differences are substantial across categories

Business Interpretation

Bike products are positioned as premium offerings and contribute most revenue per unit sold.

Possible business actions:

- Focus premium branding on bike category
- Improve bundling opportunities with accessories

Dashboard Design Features

The dashboard includes several professional dashboarding concepts:

- KPI Cards
- Consistent color palette
- Executive-level layout
- Responsive chart positioning
- Business-friendly formatting
- Clean visual hierarchy
- Interactive analytical structure

Real-World Skills Demonstrated

SQL Skills

- Aggregations
- KPI engineering
- Analytical calculations
- Data transformation
- Business metric generation

Tableau Skills

- Dashboard design
- KPI cards
- Histograms

- Bar charts
- Trend analysis
- Dashboard formatting
- Business storytelling

Data Engineering Concepts

- ETL workflow understanding
- Data warehouse layering
- Business-ready data modeling
- Reporting layer preparation

Product Analytics Skills

- Product performance analysis
- Pricing analytics
- Segment analysis
- Revenue analysis
- Product lifecycle understanding

Business Intelligence Skills

- Executive reporting
- Data storytelling
- Business interpretation
- Insight-driven analytics

Business Impact of the Dashboard

This dashboard enables businesses to:

- Monitor product performance
- Identify top revenue-generating products
- Analyze category contribution
- Improve inventory decisions
- Understand pricing behavior
- Detect slow-moving products
- Support strategic product decisions

Future Enhancements

Potential future improvements include:

- Product profitability analysis
- Margin analysis
- Predictive sales forecasting
- Product recommendation systems
- Inventory optimization
- Interactive filtering
- Drill-down product analysis

Conclusion

The Product Analytics Dashboard successfully transforms raw product transaction data into actionable business insights.

The project demonstrates:

- End-to-end analytics workflow
- Data warehouse understanding
- SQL analytics implementation
- Tableau dashboard development
- Business intelligence storytelling
- Product performance analysis

This project closely simulates real-world Product Analytics solutions used by organizations to optimize product strategy, pricing, inventory management, and revenue growth.